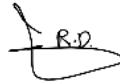


K.H.U. Faculty of Engineering

**FENS 402 ENGINEERING DESIGN PROJECT
PROGRESS REPORT**

**EMOTION RECOGNITION FROM TURKISH AUDIO IN
WORD AND SENTENCE LEVELS**



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PROGRESS:

Overview:

The Turkish Speech Emotion Recognition project uses the TurEV-DB dataset, which consists of four main emotions (angry, happy, calm, sad) at the word level. This dataset allows the project to pronounce various words in Turkish within these four different emotions (angry, happy, calm, sad) and extract the characteristics of these sounds. These extracted characteristics (Pitch, Timbre, MFCC, acoustic, bass, treble, Mel-Spectrogram, Chroma features, etc.) reveal that Turkish is different and unique from other languages. The goal is to achieve high accuracy in word prediction with models trained using these characteristics, and then to develop and finalize the project for sentence prediction. The project has demonstrated success in experimental testing (Demo, Advanced Demo) using different models, in accordance with these characteristics [1,3].

In its current state, the Demo phase of the project classifies word-based model, emotion, and characteristic matching as positive and negative. The Advanced Demo phase focuses on dividing sentences into words and understanding the overall emotional state of the sentence from the resulting words. In addition, to enable models to make predictions with higher accuracy in noisy and realistic environments, techniques such as advanced preprocessing (scaling, filling in missing values, majority voting, and adding specific weights to emotions to increase the probability of accurate prediction), dataset enhancement (adding background noise, bass, treble, and crackling), and training the models to match the features contained in the sentence (speech spaces, breath, etc.) were used.

Achievements and Comparison with the Project Schedule:

The sections where progress is being made according to the project calendar are described below.

Literature Review and Dataset Selection:

Machine learning methods for Speech Emotion Recognition have changed over time, and this change varies from language to language. We initially approached the project step-by-step, considering basic model training methods. These methods include Random Forest, Gradient Boosting, SVM, and Neural Network. These model training types are prominent in classification within the SER field. More advanced models with improved accuracy and other parameters (CatBoost, CNN, DNN, KNN, MLP, XGBoost, etc.) were examined, and their predictions on Turkish language features were superior to those of basic models. Analyses of projects using the TurEV-DB dataset for the four basic emotions (anger, happy, sad, calm) were conducted, and we subsequently tested it ourselves. The TurEV-DB dataset provided a more robust and reliable foundation in these tests [1][4].

While the sources we found indicated that deep learning methods yielded higher accuracy rates, our own developed digital signal processing, mathematical sentiment recognition, and machine learning integrated models gave us higher accuracy results. To reach this conclusion, we tested deep learning models like CNN with the TurEV-DB dataset, but since they yielded lower accuracy than our results, we will continue the process by testing various machine learning methods (Catboost, XGBoost, etc.) [5][6].

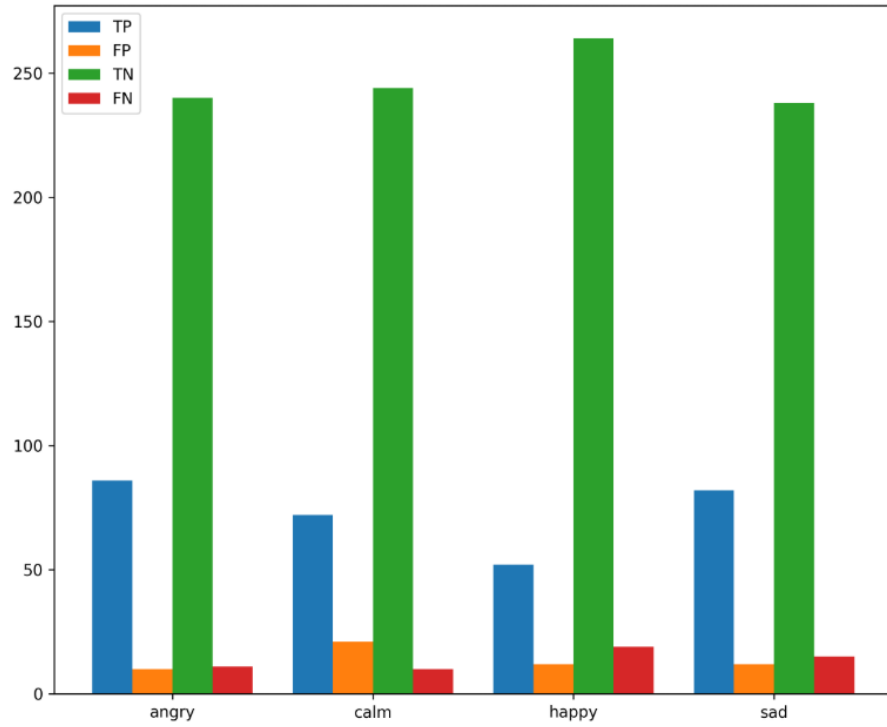


Figure 1: Class-based metrics (True Positive, False Positive, True Negative, False Negative) for the SVM model; this image is taken from the demo version of our project.

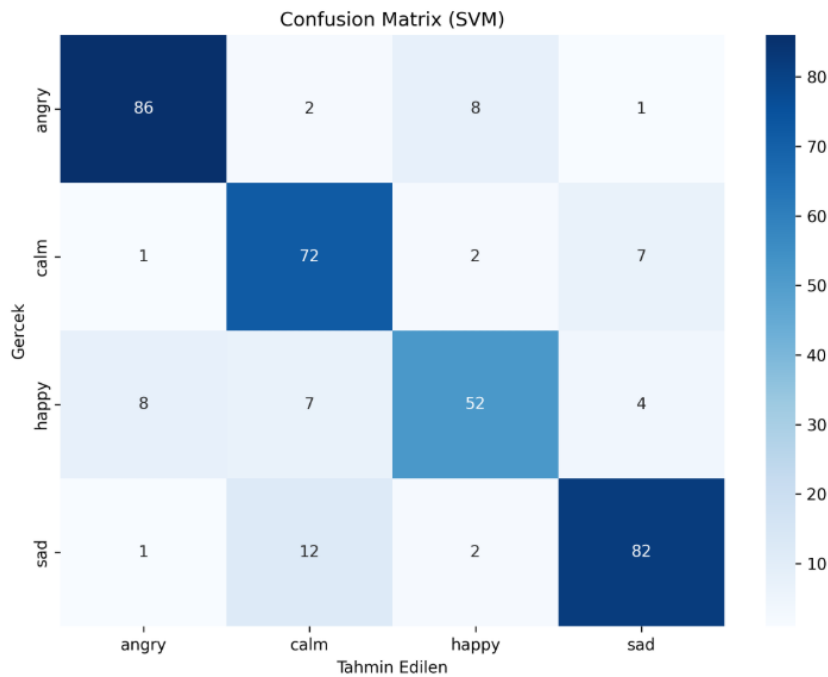


Figure 2: In the SVM model, the Confusion matrix represents the actual values horizontally and the predicted values vertically; this image is taken from the demo version of our project.

Table 1: The table shows the accuracy rates, cross-validation score, and cross-validation standard deviation of the trained models.

Model	Test Accuracy	CV Mean	CV Std.
Random Forest	0.8473	0.8408	0.0143
Gradient Boosting	0.8790	0.8653	0.0058
SVM	0.8415	0.8494	0.0115
Neural Network	0.8444	0.8429	0.0143

The phonetic structure of the Turkish language and which feature is important in which emotion in breaking down the sentence into words have been investigated, and algorithms suitable for this have been established in models and preprocessing [2]. The voice characteristics and emotion matching we obtained from the dataset and analyzed in the literature played a significant role in our preprocessing algorithm and in word detection from sentences. This increased the accuracy rate of the models obtained from the pure dataset; our work on preprocessing continues.

Dataset Preprocessing:

The Turkish Speech Emotion Recognition project uses the TurEV-DB dataset as its main source. We are continuing our email correspondence to access datasets such as BAUM-1, developed within different universities, in order to feed our trained models with more data [7]. Currently, the project continues with the dataset containing the voice recordings of individual words and their parsed features found in the TurEV-DB dataset. All outputs obtained have been acquired from this dataset. The reason we chose this dataset is that it selects the main emotions, consisting of 408 calm (neutral) voice recordings, 487 angry, 483 sad, 357 happy voice recordings and their features (pitch, timbre, jitter, shimmer, loudness, energy, etc. and minimum, maximum, std, mean, median for each feature) [8].

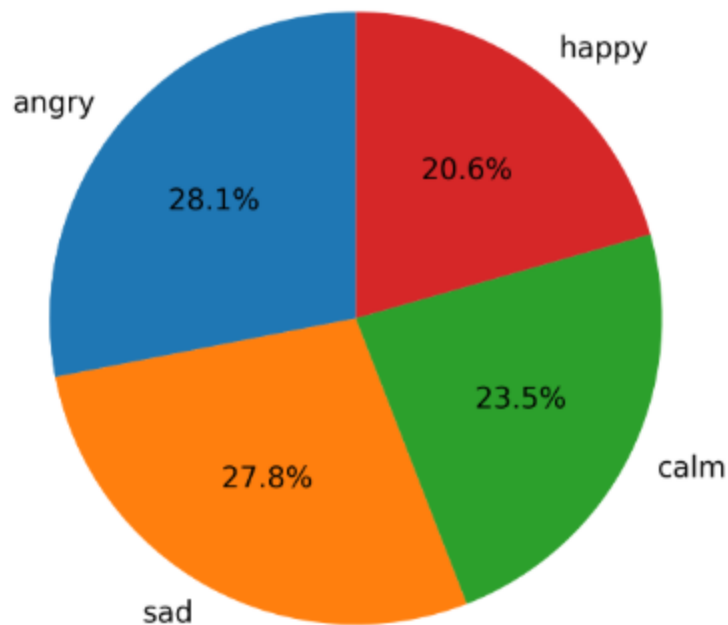


Figure 3: Pie chart showing the distribution of TurEV-DB [3] data within the dataset; this image is taken from the demo version of our project.

In the preprocessing stage, we first ensured that the features were within equal or similar ranges, which we achieved through normalization. We manually restricted the energy level to remove non-speech noises or silences, and we also developed an algorithm that discards abnormal parts using statistical values for each feature. This algorithm made it easier to capture words, but it's not entirely reliable because it discards parts of the word. Therefore, we are exploring methods using artificial intelligence to reduce noise, break down rapid speech, capture sudden bursts of energy, and remove breath or incomplete syllables. In addition to the preprocessing process, we are also researching Python libraries that will help train the models; for example, by adding external sounds, high energy, treble, and bass to our dataset to create noise, we aim to achieve a higher accuracy rate in real-time speech with a microphone.

We are progressing through the project by testing different microphones, but although the results we obtain from each microphone are similar, there are slight differences. For example, static and muffled sound from a low-quality microphone cause

high-energy signals to be captured, thus increasing the probability of predicting angry emotions. To prevent this, we are trying to find the most optimal setting, and we are also continuing our literature review on a setting that varies depending on the type of microphone. We plan to add this process to the preprocessing phase.

One of the testing phases we plan for the near future is to examine the F1 Scores (the harmonic mean of precision and recall) for four different emotions on all selected models and refine the algorithm based on the lower score. This way, even if the accuracy rate is initially misleading, we will create a more reliable model using the F1 Scores. Additionally, we intend to apply the ROC (Receiver Operating Characteristic) curve to all models. This will allow us to understand how well the models define the classes; for this, we will look at the Area Under the Curve (AUC) [9].

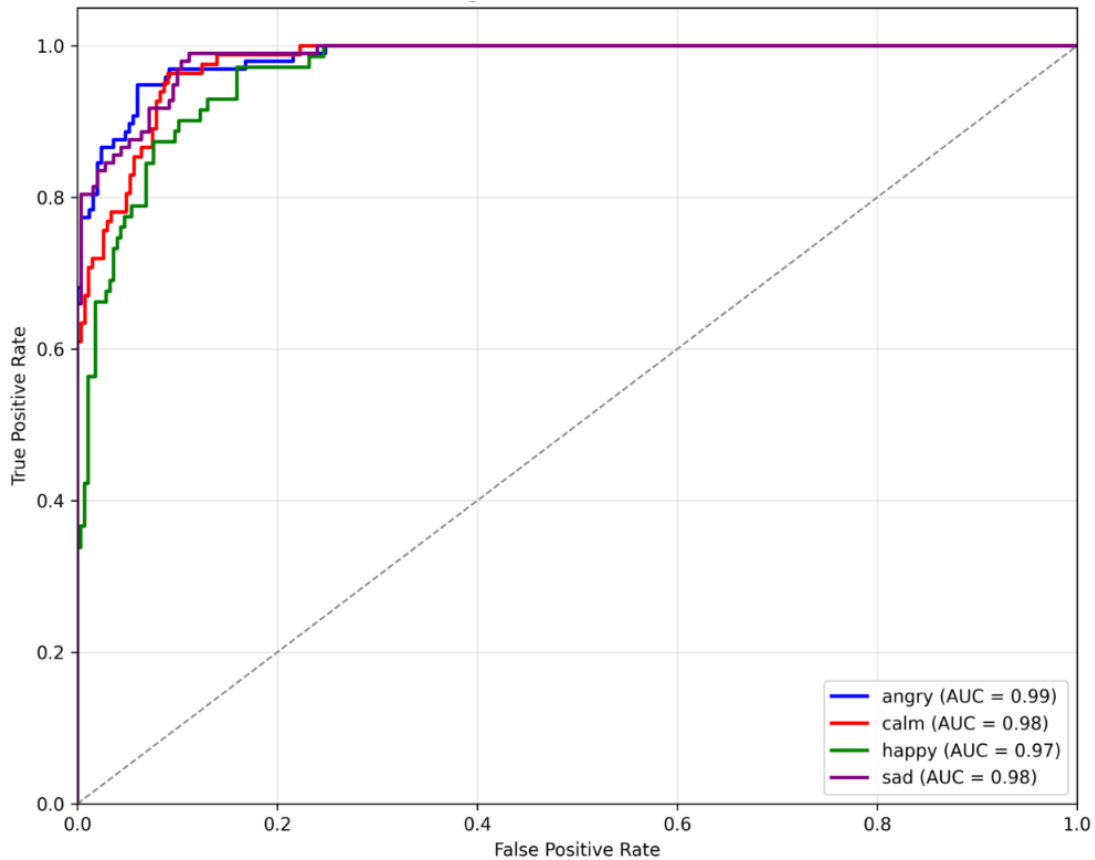


Figure 4: ROC curve and AUC scores for the SVM model; this image is taken from the demo version of our project.

Feature Extraction:

In our Turkish SER project, the Demo and Advanced Demo phases are currently complete, and testing and development are ongoing with the Advanced Demo. The Demo project focuses solely on analyzing emotions using words and training models to accurately capture specific emotions based on words. The Advanced Demo, on the other hand, predicts emotions word by word by separating the input given as a sentence.

In the demo phase, since we wanted to maintain the accuracy of the models trained with raw audio data, we used methods such as noise reduction, energy reduction, and signal-to-noise enhancement to clean the input audio. We tried to standardize the audio using constant data, thus providing an audio input close to the TurEV-DB dataset and obtaining more accurate results. To increase accuracy, in the Advanced Demo, after extracting different features from the input audio file, we applied a high-dimensional vector conversion process, making it suitable for deep learning. Additionally, to prevent inconsistencies in the data, we re-examined the normalization, bringing the values closer together without distorting the data. For example, when pitch was between 100 and 500, jitter was between 0 and 1; therefore, the models could not establish a connection between the two in the results. We brought these and similar values together on a common denominator. In the tests we obtained, we saw that models trained with machine learning outperformed deep learning, so in the coming days we will apply and test different machine learning methods (CatBoost, LightGBM, etc.). To improve the models ability to perceive processed data, we will use methods such as noise injection, combining different emotion clusters and silence with random seconds, and adding bass and treble during model training in the coming days. This will increase the models sensitivity to sound obtained from live microphone recordings. We plan to reduce the models error rate in different environments and different acoustic recordings, and the difference between the Demo and Advanced Demo proves this. While the Demo captures emotion on a word-by-word basis, it makes mistakes by predicting angry sounds as bass and treble increase and sad sounds as energy decreases. Advanced Demo, on the other hand, breaks down the incoming sentence according to its energy level and calculates the confidence rate of each prediction. This

allows it to select the emotion with the highest confidence rate as the main output, thus providing a better probability of capturing the emotion compared to the standard Demo [10,17].

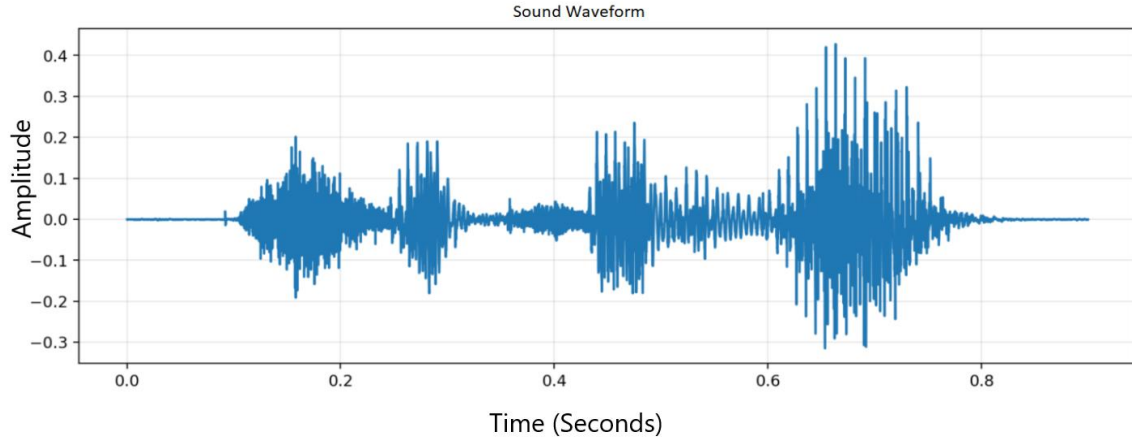


Figure 5: The sound waveform of an angry voice recording; this image is taken from the demo version of our project.

Feature Extraction and Transition to Sentence Level:

The Turkish SER Project, as indicated in the dataset, attempts to capture emotion from all angles using different parameters. Due to the unique rhythmic, temporal, and energetic characteristics of the Turkish language, emotions have their own specific definitions. For example, in Turkish, an increase in the micro-instability of sound vibrations increases the likelihood of the emotion being sad, as this occurs because energy and control decrease [18].

The Demo and Advanced Demo phases of the project utilized MFCC (Mel-Frequency Cepstral Coefficients), Mel-Spectrograms, and Chroma Features. MFCC is a voice processing technique that provides parameters for observation and measurement in voice analysis. In this way, the input is processed, compared with what the model has learned, and predictions are made. In the Demo phase of the project, voice analysis is performed using the MFCC coefficients we created. In the Advanced Demo phase, we included Mel-Spectrograms, enabling the definition of sound energy in a way that is

suitable for the human ear, increasing linearly at low frequencies and logarithmically at high frequencies. This system plays a significant role in anger and sadness thresholds by defining the energy of the incoming input as mel. Chroma Features are included in both phases; this feature adapts the sound to 12 notes and captures pitch and harmonic characteristics. By revealing different sound tones, it helps in the model's prediction.

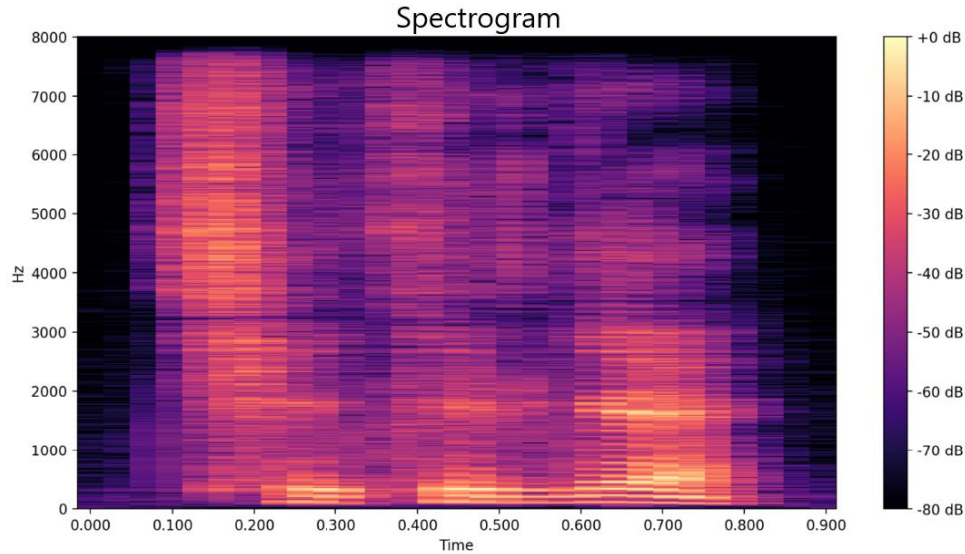


Figure 6: Spectrogram image for an angry voice recording; this image is taken from the advanced demo version of our project.

Currently, our project is having difficulty identifying the words in the input audio recording given as sentences. The algorithm works in two separate ways to capture energy and silence, but we are working on integrating an AI that will understand words semantically, or updating our existing algorithm to capture different parameters, to further improve it. In Turkish, the rapid pronunciation of words causes our system to perceive them as a single word. To prevent this, if the AI-assisted word detection and the scenario of processing and sending durations into our model fail, we will try solutions such as windowing and forced alignment. Our current system, which detects words individually and takes a weighted average based on confidence level, can switch to a weighted average based on the characteristics of the Turkish language (the weight of the emotion conveyed by the word depending on its position in the sentence), and we are working on this. This process aims to prevent the perception of a sentence as a single word or a group of words as a single word.

Model Architecture & Implementation

In this part of the project, we will describe the current state of the model, the changes in its architecture, and the integration we carried out during the transition from word-level to sentence-level.

In the early stages of the project, as seen in the demo architecture diagram, we tried to obtain outputs through a simple preprocessing operation on an audio file (Figure 7). However, as the number of models used in the project increased, access to an interface and options such as audio recording became crucial. Therefore, we developed the advanced demo and created a system that can use many models simultaneously (Figure 8).

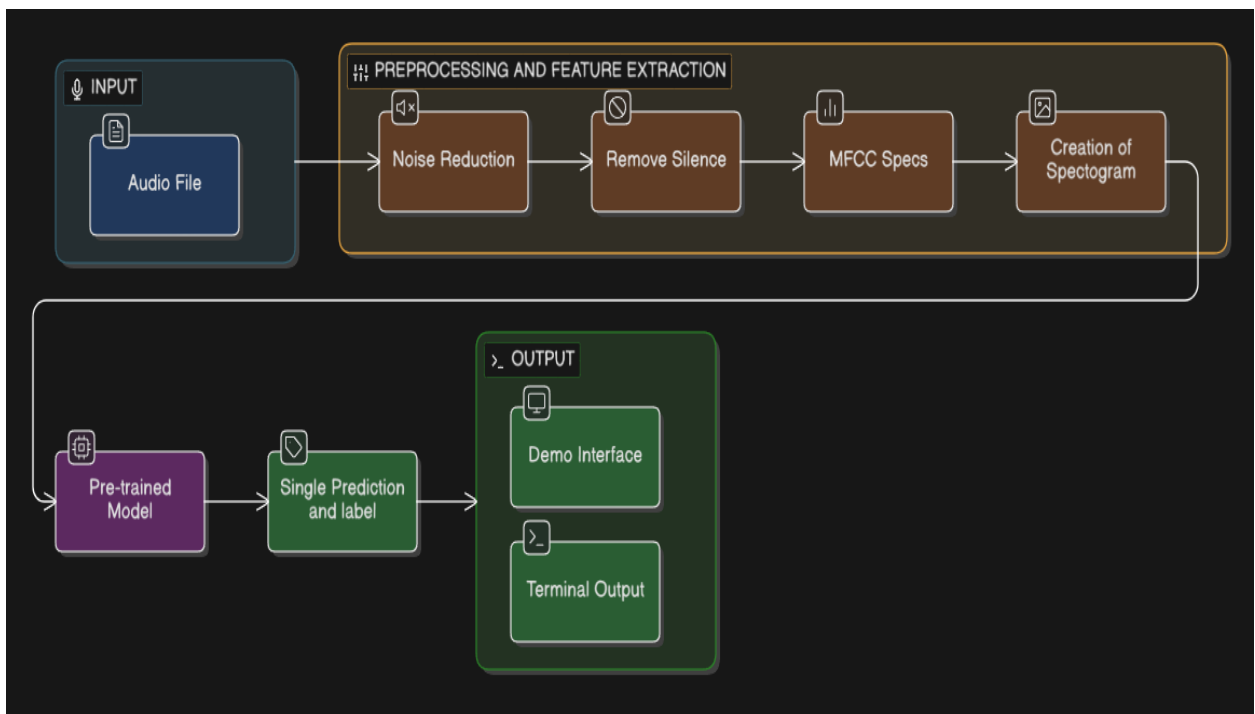


Figure 7: Demo Stage Architecture Diagram [16]

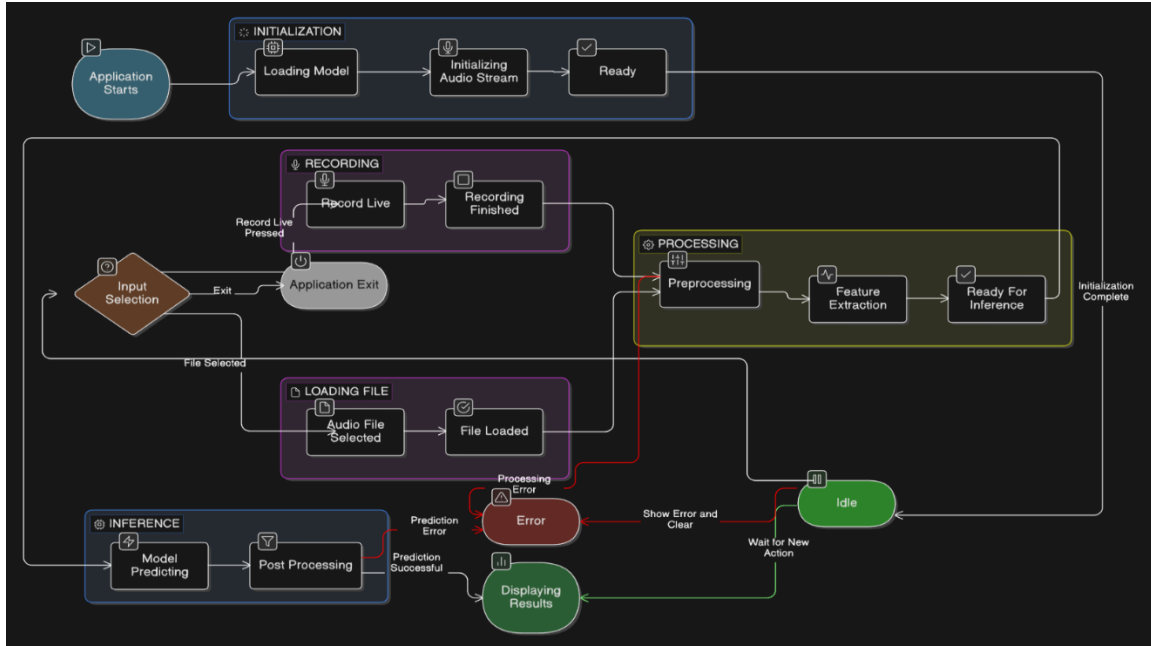


Figure 8: Advanced Demo Architecture Diagram [16]

Evaluation of the Segmentation Approach and Exploration of Alternative Methods:

In the preliminary stages of the project, we planned to use the Silence-Based Speech Activity Detection (VAD) method to understand and distinguish words in sentences, and we integrated it into the first prototypes. However, while this approach seemed logical on paper, it was realized that this separation method, which relies on energy drops, did not perform as expected in practice in a language like Turkish, which is agglutinative and where accents change dynamically. We are researching different methods to solve this problem and plan to integrate this researched solution into our project in the future. As a result of our research, we are considering using the speech-to-text model as the most logical solution. With this model, we will be able to understand the words in the sentence and determine which word was spoken between which seconds, thus successfully distinguishing the words. This is our plan.

Increasing Model Diversity and Pool:

In the initial prototypes of our project, we had a trained model, but it wasn't very accurate. To maximize the robustness of our system and prevent overfitting problems, we planned to use multiple models instead of a single one. Currently, As we can see from the Figure 9 the project includes Random Forest, Gradient Boosting, CNN-LSTM, and XGBoost models. When we first designed the project, the CNN-LSTM model seemed the most logical on paper, but in practice, the results weren't what we expected. Therefore, in addition to the CNN-LSTM model, we trained three more models, bringing our current model count to four. We plan to increase the number of models further in the future.

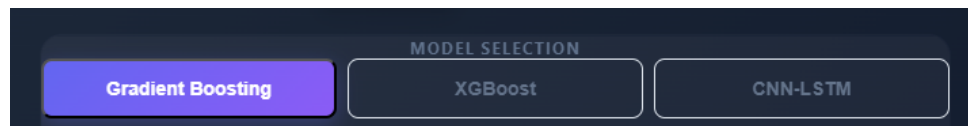


Figure 9: Model Selection UI

Establishing the Majority/Weighted Voting Mechanism:

In the sentiment prediction phase, which we had planned even in the draft stage of the project, we successfully integrated majority voting in our project as we can see from the Figure 10, where multiple models score each word to determine its sentiment. Thanks to this integration, we are getting closer to a more accurate result by having multiple models score each word. For example, in the sentence "The weather is so nice today, we definitely need to have a picnic," we score the word "weather" individually with XGBoost, CNN-LSTM, Gradient Boosting, and Random Forest models, and then we determine which sentiment that word corresponds to based on the combined decision of all of them. In the later stages of the process, we plan to make this mechanism a more detailed scoring and decision-making mechanism based on the sentence's conjunctions, verbs, subjects, etc.

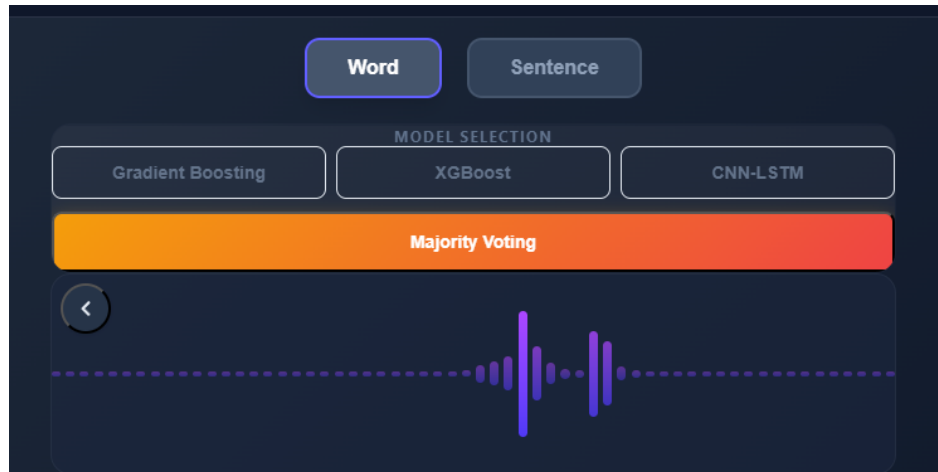


Figure 10: Experimental Results & Evaluation UI

This part of the project reports on the results of the architectural changes we made and the ongoing model testing processes.

VAD Test Results and STT's Potential:

In the initial experiments conducted on the system prototype, it was observed that the VAD algorithm could not distinguish words from each other in fast speech or mistakenly split a word into two during breathing pauses in sentences. Due to this issue, alternative methods were investigated, and it was decided that the Speech-to-Text model was the most logical word separation method. We anticipate that this model will significantly increase the accuracy of word extraction.

Examining and Comparing Model Performances:

The training processes for the 3-4 different models we have developed are still ongoing. We have begun comparing the models' success not only using the accuracy metric but also on the confusion matrix. This allows us to understand which emotions the models confuse with each other.

Majority Voting Integration Pre-Tests:

We are getting promising results in sentence-based sentiment prediction because we have integrated a voting algorithm into our system. Currently, the system predicts the sentiment of each word individually, assigning equal scores to all words through majority voting. Based on our initial tests, we believe that in later stages we should also consider the sentiment weighting of words to achieve more accurate word-based predictions. Therefore, we are researching a more advanced weighted voting algorithm that reduces the influence of neutral words on the prediction process and focuses on the impact of words with the actual sentiment affecting the sentence's sentiment, and we plan to integrate this algorithm into our system.

Shortcomings, Challenges, and Precautions:

Despite the gains and progress achieved in the demo and advanced demo, certain problems were encountered throughout the project.

One of these was the inadequacy of the Voice Activity Deduction (VAD) application for word separation. The language structure, including intonations like the elision in Turkish, led to the combined pronunciation of some words. This resulted in words not being sharply separated. To overcome this problem, we aim to include an AI-powered Speech-to-Text structure in the later stages of the project to achieve sharper word understanding.

Another problem was the varying sound sensitivities of different microphones in the project. Despite all the preprocessing methods applied, microphone differences still affected the results. To overcome this problem, we focused on minimizing variations between different microphones by including new sounds containing artificial noise in the dataset and using robust models.

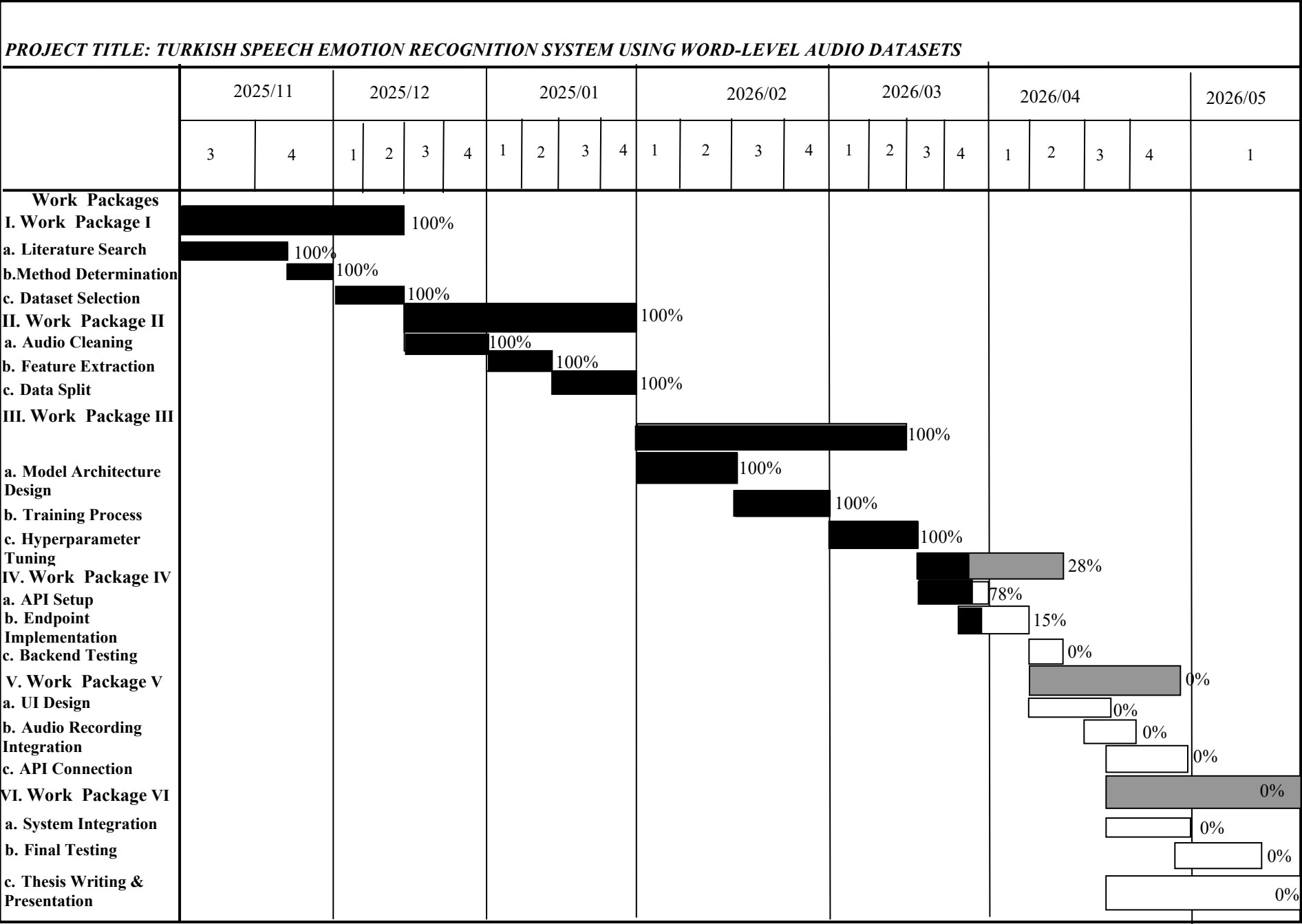
Future Work: Transition to Sentence-Level Analysis

In our ongoing work, we decided to make changes to the sentence analysis section and include an AI-powered Speech-to-Text structure. This approach will help us obtain more consistent results on the path from word analysis to sentence analysis. We anticipate that sharper word distinctions will also improve the performance of the models.

Furthermore, we plan to add new models to ensure model diversity and increase consistency. We believe that we will obtain more consistent results with methods such as model diversity and weighted voting, and we are continuing our work to select the models we will continue to use in the future.

To ensure that we obtain different results in different test environments and that the project is usable in the real world, we are focusing on preprocessing methods for different microphones and diversifying datasets to make the project generally user-friendly.

Project Calender:



EXPENDITURES:

	Work Packages				
	I	II	III	IV	TOTAL
Equipment	-	-	-	-	-
Consumable goods	-	-	-	-	-
Publications/Software	-	-	-	-	-
Transportation	-	-	-	-	-
Services	-	-	-	-	-
TOTAL	-	-	-	-	-

Equipment:

Mid-range or high-end personal computers: 3 laptops and 3 desktop computers belonging to team members.

External Microphone (Optional external low, medium, high-quality microphone): Microphones for cleaner recordings during the testing process and to achieve maximum accuracy with different quality microphones.

Consumable goods:

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Publications/Software:

Python [19], Streamlit[20], Librosa[21], Scikit-Learn[22], Pandas[23], NumPy[24], Matplotlib & Seaborn[25,26], SciPy[27], SoundFile[28], Plotly[29], Support Vector Machines (SVM) [30], Gradient Boosting [31], Multi-layer Perceptron (MLP) [32], Principal Component Analysis (PCA) [33], GridSearchCV, Pickle.

Transportation:

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Services:

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